

[Secure Exit](#)**SIMATS**
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO4-AT2-Lab Practice

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QUESTIONS

Q1Java Lab Practice – Applets,
AWT Components and Layout
Managers

100 marks

1/1 answered

Q1

Java Lab Practice – Applets, AWT Components and Layout Managers

100
marks

Question 1 – Applet Life Cycle

Develop a Java Applet to demonstrate the Applet Life Cycle methods: `init()`, `start()`, `paint()`, `stop()`, and `destroy()`.

Display a suitable message in the applet window whenever each life-cycle method is invoked. Observe the sequence of execution by running the applet and interacting with it.

Expected Outcome:

Students should understand the execution sequence and purpose of each Applet life-cycle method.

Question 2 – Generating Graphic Images Using Applet

Develop a Java Applet that uses the Graphics class to generate a simple graphical scene.

The applet should draw:

A house

Sun

Trees

Road

Clouds

Use methods such as `drawLine()`, `drawRect()`, `drawOval()`, `fillRect()`, `fillOval()`, and appropriate colors.

Expected Outcome:

Students should demonstrate the use of the Graphics class for drawing shapes and images.

Question 3 – Calculator Application Using Applet Parameters

Develop a Calculator Applet that accepts two numbers and an arithmetic operator through HTML/Applet parameters.

The applet should perform:

Addition

Subtraction

Multiplication

Division

Use the `getParameter()` method to retrieve the values passed to the applet.

Expected Outcome:

Students should understand how parameters are passed from an HTML page to an Applet and processed in Java.

Question 4 – Student Profile Using AWT Components

Develop a Student Profile Application using Java AWT components such as:

Label

TextField

Checkbox

CheckboxGroup

Choice

TextArea

Button

List

The application should accept student details such as:

Name

Register Number

Gender

Department

Year

Skills/Programming Languages

When the Generate Report button is clicked, display the entered student information in a List or TextArea.

Expected Outcome:

Students should demonstrate the use of multiple AWT components and event handling.

Question 5 – GridLayout with Event Handling

Develop a Java AWT application using GridLayout to create a simple form or calculator interface.

Arrange the components in a grid and implement appropriate event handling using ActionListener.

For example, create a calculator with buttons:

7 8 9 +

4 5 6 -

1 2 3 *

0 = C /

Perform the corresponding operation when a button is clicked.

Expected Outcome:

Students should understand GridLayout and button-based event handling.

Question 6 – GridBagLayout with Event Handling

Develop a Java AWT Student Registration Form using GridBagLayout.

The form should contain:

Name

Register Number

Department

Email

Gender

Year

Submit Button

Reset Button

Use GridBagConstraints to properly position and align the components.

Implement event handling for the Submit and Reset buttons.

Expected Outcome:

Students should understand flexible component positioning using GridBagLayout and event handling.

Question 7 – BorderLayout with Event Handling

Develop a Java AWT application using BorderLayout.

Divide the window into:

NORTH – Header

SOUTH – Buttons

EAST – Options

WEST – Menu

CENTER – Display area

Add buttons in the interface and implement event handling so that clicking different buttons changes the content displayed in the center area.

Expected Outcome:

Students should understand the five regions of BorderLayout and event-driven programming.

Question 8 – CardLayout with Event Handling

Develop a Java AWT application using CardLayout to create a multi-page interface.

Create at least three cards:

Card 1: Student Information

Card 2: Academic Information

Card 3: Confirmation/Report

Provide Next, Previous, and Submit buttons to navigate between the cards.

Implement event handling to switch between cards using CardLayout.

Expected Outcome:

Students should understand how CardLayout can be

used to create multiple screens within a single window.

Lab Practice Evaluation Focus

Students will be evaluated based on:

Correct implementation of the required Java concepts.

Proper use of AWT components.

Correct implementation of event handling.

Appropriate use of layout managers.

User-friendly interface design.

Successful compilation and execution.

Ability to explain the program and output during the lab examination.

✓ Submitted Files



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Awaiting faculty evaluation